2.58 Graph Sketching: Reciprocal Functions

Worked Solutions

1 a)

Asymptotes: $x = -2$, $x = 2$.

$$\text{Minimum of } f(x) \text{ at } (0, -4) \text{ becomes local maximum of } \frac{1}{f(x)} \text{ at } \left(0, -\frac{1}{4}\right).$$

As $x \to -\infty$, $\frac{1}{f(x)} \to 0^+$; as $x \to \infty$, $\frac{1}{f(x)} \to 0^+$.

$$x \to -2^- : \frac{1}{f(x)} \to \infty, \qquad x \to -2^+ : \frac{1}{f(x)} \to -\infty,$$
$$x \to 2^- : \frac{1}{f(x)} \to -\infty, \qquad x \to 2^+ : \frac{1}{f(x)} \to \infty.$$

1 b)

Asymptotes: $x = 0$, $x = 3$.

$$\text{Minimum of } f(x) \text{ at } \left(\frac{3}{2}, -\frac{9}{4}\right) \text{ becomes local maximum of } \frac{1}{f(x)} \text{ at } \left(\frac{3}{2}, -\frac{4}{9}\right).$$

As $x \to -\infty$, $\frac{1}{f(x)} \to 0^+$; as $x \to \infty$, $\frac{1}{f(x)} \to 0^+$.

$$x \to 0^- : \frac{1}{f(x)} \to \infty, \qquad x \to 0^+ : \frac{1}{f(x)} \to -\infty,$$
$$x \to 3^- : \frac{1}{f(x)} \to -\infty, \quad x \to 3^+ : \frac{1}{f(x)} \to \infty.$$

1 c)

Asymptotes: $x = -2$, $x = 1$.

$$\text{Maximum of } f(x) \text{ at } \left(-\frac{1}{2}, \frac{9}{4}\right) \text{ becomes local minimum of } \frac{1}{f(x)} \text{ at } \left(-\frac{1}{2}, \frac{4}{9}\right).$$

As $x \to -\infty$, $\frac{1}{f(x)} \to 0^-$; as $x \to \infty$, $\frac{1}{f(x)} \to 0^-$.

$$x \to -2^- : \frac{1}{f(x)} \to -\infty, \quad x \to -2^+ : \frac{1}{f(x)} \to \infty,$$
$$x \to 1^- : \frac{1}{f(x)} \to \infty, \qquad x \to 1^+ : \frac{1}{f(x)} \to -\infty.$$

2 a)

When $x = 0$, $y = -1$. Asymptotes: $x = \frac{1}{3}$, $y = 0$.

2 b)

When $x = 0$, $y = -\frac{1}{12}$. Asymptotes: $x = -3$, $x = 4$, $y = 0$. The minimum of $y = (x-4)(x+3)$ is at $x = \frac{1}{2}$. Hence the maximum of $y = \frac{1}{(x-4)(x+3)}$ is at $\left(\frac{1}{2}, -\frac{4}{49}\right)$.

2 c)

When $x = 0$, $y = -\frac{1}{2}$. Asymptotes: $x = -2$, $x = \frac{1}{2}$, $y = 0$. The minimum of $y = 2x^2 + 3x - 2$ is at $x = -\frac{3}{4}$. Hence the maximum of $y = \frac{1}{2x^2+3x-2}$ is at $\left(-\frac{3}{4}, -\frac{8}{25}\right)$.

2 d)

When $x = 0$, $y = \frac{1}{4}$. Asymptotes: $x = -1$, $x = 4$, $y = 0$.

Maximum of $y = 4 + 3x - x^2$ is at $x = \dfrac{3}{2}$, so the minimum of $y = \dfrac{1}{4 + 3x - x^2}$ is at $\left(\dfrac{3}{2}, \dfrac{4}{25}\right)$.

3 a)

Asymptotes: $x = -2$, $x = -1$, and $x = 2$. When $x = 0$, $y = -\frac{1}{4}$. As $x \to -\infty$, $\frac{1}{f(x)} \to 0^-$; as $x \to \infty$, $\frac{1}{f(x)} \to 0^+$.

$$x \to -2^- : \frac{1}{f(x)} \to -\infty, \quad x \to -2^+ : \frac{1}{f(x)} \to \infty,$$
$$x \to -1^- : \frac{1}{f(x)} \to \infty, \quad x \to -1^+ : \frac{1}{f(x)} \to -\infty,$$
$$x \to 2^- : \frac{1}{f(x)} \to -\infty, \quad x \to 2^+ : \frac{1}{f(x)} \to \infty.$$

3 b)

Asymptotes: $x = -1$, $x = 0$, and $x = 2$. As $x \to -\infty$, $\frac{1}{f(x)} \to 0^+$; as $x \to \infty$, $\frac{1}{f(x)} \to 0^-$.

$$x \to -1^- : \frac{1}{f(x)} \to \infty, \quad x \to -1^+ : \frac{1}{f(x)} \to -\infty,$$
$$x \to 0^- : \frac{1}{f(x)} \to -\infty, \quad x \to 0^+ : \frac{1}{f(x)} \to \infty,$$
$$x \to 2^- : \frac{1}{f(x)} \to \infty, \quad x \to 2^+ : \frac{1}{f(x)} \to -\infty.$$

3 c)

Asymptotes: $x = -1$ and $x = 0$. As $x \to -\infty$, $\frac{1}{f(x)} \to 0^-$; as $x \to \infty$, $\frac{1}{f(x)} \to 0^+$.

$$x \to -1^- : \frac{1}{f(x)} \to \infty, \quad x \to -1^+ : \frac{1}{f(x)} \to -\infty,$$
$$x \to 0^- : \frac{1}{f(x)} \to -\infty, \quad x \to 0^+ : \frac{1}{f(x)} \to \infty.$$

4 a)

Asymptotes: $x = 0$, $x = 1$, $x = 2$, $y = 0$. As $x \to -\infty$, $\frac{1}{f(x)} \to 0^-$; as $x \to \infty$, $\frac{1}{f(x)} \to 0^+$.

$$x \to 0^- : \frac{1}{f(x)} \to -\infty, \quad x \to 0^+ : \frac{1}{f(x)} \to \infty,$$
$$x \to 1^- : \frac{1}{f(x)} \to \infty, \quad x \to 1^+ : \frac{1}{f(x)} \to -\infty,$$
$$x \to 2^- : \frac{1}{f(x)} \to -\infty, \quad x \to 2^+ : \frac{1}{f(x)} \to \infty.$$

4 b)

$$2x^3 - 9x^2 + 3x + 4 = (2x+1)(x-1)(x-4).$$

Asymptotes: $x = -\frac{1}{2}$, $x = 1$, $x = 4$, $y = 0$. When $x = 0$, $y = \frac{1}{4}$. As $x \to -\infty$, $\frac{1}{f(x)} \to 0^-$; as $x \to \infty$, $\frac{1}{f(x)} \to 0^+$.

$$x \to -\frac{1}{2}^- : \frac{1}{f(x)} \to -\infty, \quad x \to -\frac{1}{2}^+ : \frac{1}{f(x)} \to \infty,$$
$$x \to 1^- : \frac{1}{f(x)} \to \infty, \qquad x \to 1^+ : \frac{1}{f(x)} \to -\infty,$$
$$x \to 4^- : \frac{1}{f(x)} \to -\infty, \qquad x \to 4^+ : \frac{1}{f(x)} \to \infty.$$

4 c)

$$8 - 2x - 5x^2 - x^3 = -(x-1)(x+2)(x+4).$$

Asymptotes: $x = -4$, $x = -2$, $x = 1$, $y = 0$. When $x = 0$, $y = \frac{1}{8}$. As $x \to -\infty$, $\frac{1}{f(x)} \to 0^+$; as $x \to \infty$, $\frac{1}{f(x)} \to 0^-$.

$$x \to -4^- : \frac{1}{f(x)} \to \infty, \qquad x \to -4^+ : \frac{1}{f(x)} \to -\infty,$$
$$x \to -2^- : \frac{1}{f(x)} \to -\infty, \quad x \to -2^+ : \frac{1}{f(x)} \to \infty,$$
$$x \to 1^- : \frac{1}{f(x)} \to \infty, \qquad x \to 1^+ : \frac{1}{f(x)} \to -\infty.$$

4 d)

$$x^3 + x^2 = x^2(x+1).$$

Asymptotes: $x = -1$, $x = 0$, $y = 0$. When $x = 0$, $y = 1$. As $x \to -\infty$, $\frac{1}{f(x)} \to 0^-$; as $x \to \infty$, $\frac{1}{f(x)} \to 0^+$.

$$x \to -1^- : \frac{1}{f(x)} \to -\infty, \quad x \to -1^+ : \frac{1}{f(x)} \to \infty,$$
$$x \to 0^- : \frac{1}{f(x)} \to \infty, \qquad x \to 0^+ : \frac{1}{f(x)} \to \infty.$$

5 a)

$$y = \frac{1}{k(x-1)(x-2)}.$$

Given $\left(\frac{3}{2}, -\frac{2}{5}\right)$, then

$$-\frac{2}{5} = \frac{1}{k\left(\frac{3}{2}-1\right)\left(\frac{3}{2}-2\right)} \quad \therefore \quad k = 10.$$

So

$$10(x-1)(x-2) = 10x^2 - 30x + 20, \qquad y = \frac{1}{10x^2 - 30x + 20}.$$
$$\therefore a = 10, \quad b = -30, \quad c = 20.$$

5 b)

$$y = \frac{1}{k(x+1)(x-3)}.$$

Given $(1, \frac{1}{8})$, then

$$\frac{1}{8} = \frac{1}{k(1+1)(1-3)} \quad \therefore \quad k = -2.$$

So

$$-2(x+1)(x-3) = -2x^2 + 4x + 6, \qquad y = \frac{1}{-2x^2+4x+6}.$$
$$\therefore a = -2, \quad b = 4, \quad c = 6.$$

6)

$$y = \frac{1}{k(x+1)(x-1)(x-2)}.$$

Given $(0, \frac{1}{2})$, then

$$\frac{1}{2} = \frac{1}{k(0+1)(0-1)(0-2)} \quad \therefore \quad k = 1.$$

So

$$(x+1)(x-1)(x-2) = x^3 - 2x^2 - x + 2, \qquad y = \frac{1}{x^3-2x^2-x+2}.$$
$$\therefore a = 1, \quad b = -2, \quad c = -1, \quad d = 2.$$

7)

$$\frac{1}{f(x)} = \frac{x^2-4x-5}{x^2+4x} = \frac{(x+1)(x-5)}{x(x+4)}.$$

Asymptotes: $y = 1$, $x = 0$, $x = -4$.

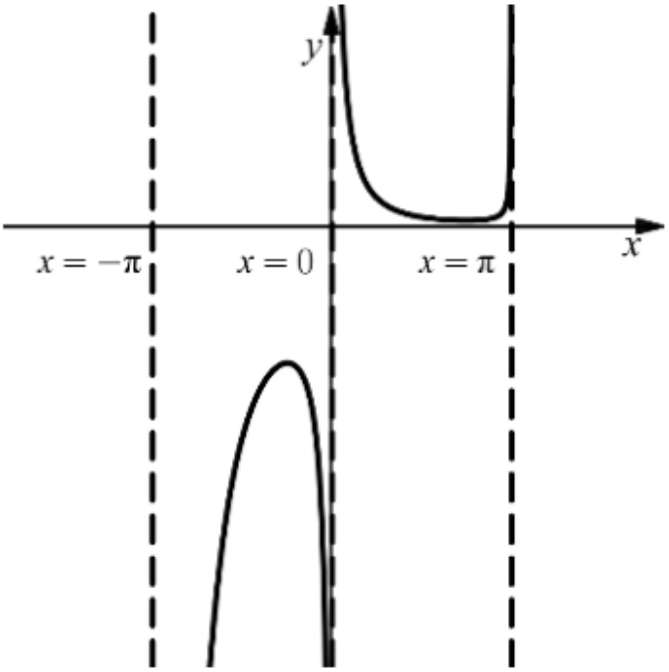
$$x^2 - 4x - 5 = 0 \quad \Rightarrow \quad x = 5, \ x = -1,$$

so the intercepts are $(5, 0)$ and $(-1, 0)$. There is no vertical intercept. As $x \to \infty$, $y < 1$; as $x \to -\infty$, $y > 1$.

$$x \to -4^- : y \to +\infty, \quad x \to -4^+ : y \to -\infty, \quad x \to 0^- : y \to +\infty, \quad x \to 0^+ : y \to -\infty.$$

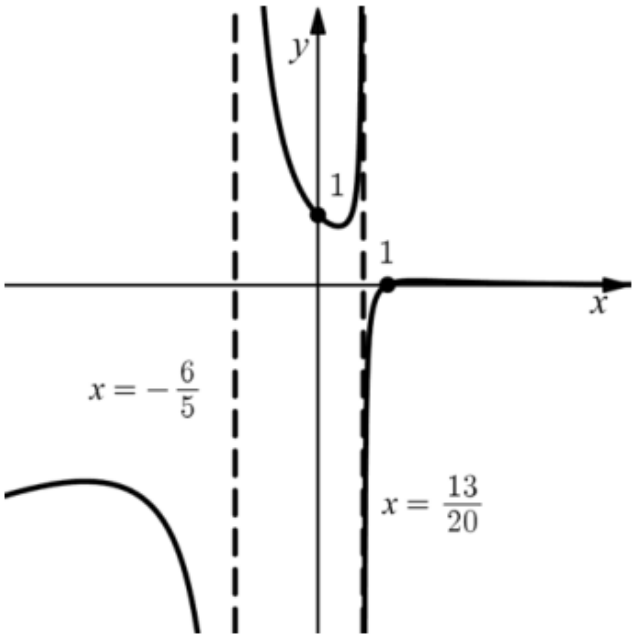
8 a)

Sketch of the reciprocal curve shown in the printed worked solution.



8 b)

Sketch of the reciprocal curve shown in the printed worked solution.



9)

Asymptotes at $x = 0$ and $x = \frac{1}{2}$.

$$\ln(bx) = 0 \quad \Rightarrow \quad bx = 1 \quad \Rightarrow \quad x = \frac{1}{b} \quad \therefore \quad b = 2.$$

$$y = \frac{1}{ax\ln(2x)} \quad \Rightarrow \quad -\frac{e}{4} = \frac{1}{a\left(\frac{1}{2e}\right)\ln\left(\frac{2}{2e}\right)} \quad \Rightarrow \quad -\frac{e}{4} = \frac{2e}{-a} \quad \therefore \quad a = 8.$$

$$\therefore a = 8 \text{ and } b = 2.$$